



The University of Faisalabad

Course Code	Course Title	Credit Hours	Contact Hrs./Week	Total Lectures Allocated
STAT-311	Probability & Statistics	3(3-0)	3(3-0)	45

Discipline	Semester no	Commencement of Semester	Mid Semester Examination	Semester Termination Date
BSCS/BSSE/BSIT/BSAI	III	11-09-2023		

BRIEF COURSE DESCRIPTION

This course includes the terminology, basic tools and concepts of statistical methods, as well as the current software in the area, to be able to communicate with analysts and other experts in the field.

Instructor Name: Muhammad Zohaib	Email: lecturer.nbs.994@tuf.edu.pk	Contact No: 0306-7172707
---	---	---------------------------------

Intended Learning Objectives

By the end of course students will learn;

- Basic tools and concepts of statistical methods.
- To be able to communicate with analysts and other experts in field.
- Apply appropriate statistical techniques for managing common types of data; use various software packages for statistical analysis and data management

Intended learning Outcomes

- To provide the students with the necessary concepts of statistics to enable them to realize a research project in the field of Computer.
- It involves selection of appropriate statistical techniques; for managing common types of data; use various software packages for statistical analysis.
- Interpret the results of statistical analyses and critically evaluate the use of statistics in the field of Computer Sciences.
- Communicate effectively with statisticians, in writing and orally through presentation of results of statistical analyses.

Contents:

Week	Lecture	Title of Chapter	Sub Heading	Contents
1	1	Introduction	Meaning of Statistics	Definition of Statistics• Descriptive and Inferential Statistics• Population and Sample• Parameter and Statistic • Data and its types
1	2		Observations and Variables	Variable and its types• Discrete and Continuous Variables
1	3		Observations and Variables	Measurement scales
2	4		Collection of Data	Collection of Primary Data• Collection of secondary Data
2	5	Presentation of Data	Frequency Distribution	Definition of Classification and Tabulation •Construction of discrete or Ungrouped frequency distribution • Relative Frequency• Percentage Frequency • Cumulative Frequency
2	6		Frequency Distribution	Construction of Grouped frequency distribution • Class Limit and Class Boundaries • Class Mark • Class Width and Class Interval • Stem and Leaf Display
3	7		Diagrams	Difference between Diagrams and Graphs •Simple Bar Chart • Multiple Bar Chart • Component Bar Chart • Pie Chart
3	8		Graphs	Histogram • Frequency Polygon • Frequency Curve • Cumulative Frequency Polygon or Ogive
3	9	Measures of Average	Mean	Arithmetic mean• Properties of A.M• Merits and demerits of A.M, Merits and demerits of Median• Numerical
4	10		Median	Merits and demerits of Median• Numerical
4	11		Mode	Merits and demerits of Mode • Numerical
4	12		Quartiles	Numerical
5	13		Exercise	Numerical
5	14		Exercise	Numerical
5	15	Quiz	Quiz	Quiz
6	16	Measure of dispersion	Introduction	Definition of Measures of Dispersion •Absolute and Relative Measures of Dispersion
6	17		The Range	Range • Coefficient of Range • Interquartile Range • Semi interquartile Range •Coefficient of Quartile Deviation
6	18		The Variance and Standard Deviation	Variance and standard deviation for ungrouped data • Interpretation of variance
7	19		The Variance and Standard Deviation	Variance and standard deviation for grouped data• Interpretation of variance• Coefficient of variation
7	20		Skewness	Skewness • Positive Skewness • Negative Skewness

7	21		Kurtosis	Lepto-Kurtic • Meso-Kurtic • Platy-Kurtic
8	22		Exercise	Numerical
8	23		Exercise	Numerical
8	24		Exercise	Numerical
10	25	Probability	Introduction	Definitions: •Experiment • Random Experiment • Sample Space • Events • Mutually Exclusive Events • Exhaustive Events• Equally Likely Events • Dependent and Independent Events.
10	26		Definitions of Probability	Classical definition of probability its numerical
10	27		Laws of Probability	Numerical
11	28		Laws of Probability	Numerical
11	29		Exercise	Numerical
11	30		Exercise	Numerical
12	31	Quiz	Quiz	Quiz
12	32	Discrete probability distributions	Binomial Distribution	Introduction • Properties of binomial distribution
12	33		Binomial Distribution	Numerical
13	34		Poisson Distribution	Introduction •Properties of Poisson distribution
13	35		Poisson Distribution	Numerical
13	36	Continuous probability distribution	Normal Distribution	Introduction and Properties of Normal Distribution • Standard Normal Distribution
14	37		Normal Distribution	Numerical
14	38	Presentation	Presentation	Presentation

14	39	Regression and Correlation	Introduction	Introduction • Deterministic and Probabilistic Relation • Scatter Diagram • Properties of Least Square Regression Line • Standard Error of Estimates
15	40		Simple Linear Regression	Numerical
15	41		Coefficient of Determination	Concept of Coefficient of Determination and Numerical
15	42		Correlation	Introduction • Types of correlation • Properties of r • Correlation and Causation
16	43	Statistical Inference Estimation	Estimation	Introduction • Basic Concepts of Estimation
16	44		Estimation	Numerical
16	45	Testing of hypothesis	Introduction	Introduction • Basic Concepts of Testing of Hypothesis • Numerical
17				

Assessment

Assignments 1: Basic Concepts of Statistics	Tentative Date of Submission:25-09-2023
Assignments 2: Concepts of Descriptive Statistics	Tentative Date of Submission:00-00-2023
Assignments 3: Probability	Tentative Date of Submission:00-00-2023
Assignments 4: Regression and Correlation	Tentative Date of Submission:00-00-2023
Presentation 1: Numerical related to Probability, Regression and Correlation	Tentative Date of Submission:00-00-2023
Quiz1: Discrete and continuous distribution	Tentative Date of Submission:00-00-2023
Quiz2: Statistical inference	Tentative Date of Submission:00-00-2023

Recommended Text Books

Sr.	Title	Authors
1	Probability & Statistics for Engineers & Scientists	Ronald Walpole, Myers, Myers
2	Introduction to statistical theory I	Prof. Shar Muhammad Chaudhry
3	Introduction to statistical theory II	Prof. Shar Muhammad Chaudhry

Assessment Scheme:

Internal Assessment – 20%

- Quiz:
- Assignment:

- Presentation:

Mid Semester Examination - 30%

Final Semester Examination - 50%

	Name	Designation	Date	Signature
Instructor	Muhammad Zohaib	Lecturer	11 September 2023	
Head of School				

Note: - The above course outline covers all contents of the subject as prescribed by HEC and respective accreditation Council.